VASANTH T M

Full-Stack Developer | Applied AI & IoT Systems

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PROFESSIONAL SUMMARY

B.Tech Artificial Intelligence and Data Science student and multi-hackathon-winning AI engineer with hands-on experience across IoT systems, full-stack web development, and Agentic AI workflows. Shipped applied projects spanning smart traffic monitoring, accident-alert systems, and an AI-driven healthcare validation platform, backed by a hackathon win at MobilityX 2.0. Vice Secretary of the IEEE Student Branch, combining backend development strength with cross-functional leadership.

EDUCATION

Bannari Amman Institute of Technology 2024 – 2028
B.Tech, Artificial Intelligence and Data Science
CGPA: 8.1 (current, 5th semester)

Vidhyaa Vikas MHS School 2021 – 2024
Higher Secondary Education
HSC: 92% | SSLC: 78.6%

TECHNICAL SKILLS

Programming Languages: C, C++, Python

Web Technologies: HTML, CSS, JavaScript, React

Backend & Databases: FastAPI, PostgreSQL

AI & Automation: Agentic AI workflows, API integration

Tools & Platforms: Git, Redis, Celery

HACKATHONS & ACHIEVEMENTS

Winner, MOBILITYX 2.0, *AI Clashers*
Cash Prize; national-level 24-hour innovation hackathon with team AI Clashers, presented before EV industry experts, researchers, and innovators.

2nd Runner-Up, HackXelerate'26
2nd Runner-Up among 600+ participants; delivered an AI solution through multiple evaluation rounds.

Vice Secretary, IEEE Student Branch
Coordinated technical events and student initiatives.

Outreach & External Affairs Executive, IEEE RAS
Led outreach efforts and external partnerships for the Robotics & Automation Society chapter.

Career Guidance Mentor, Government Higher Secondary Students
Mentored students on career pathways in technology and higher education.

KEY PROJECTS

Smart Traffic Monitoring & Accident Alert System, *ESP32, IR Sensors, GPS, IoT*

- Built an IoT-based system using ESP32, IR sensors, and flame detection to monitor traffic conditions in real time.
- Integrated GPS and automated SOS alerts to enable rapid accident emergency response.
- Designed the system for traffic control coordination alongside emergency alerting.

Agentic AI Healthcare Validation Platform, *React, FastAPI, Celery, Redis, PostgreSQL*

- Developed a full-stack Agentic AI platform for automated healthcare data validation.
- Implemented automated data extraction pipelines with asynchronous task processing via Celery and Redis.

- Built API-based verification workflows backed by a PostgreSQL data layer.

Hostel Booking Web Application, *React, FastAPI, PostgreSQL*

- Built a full-stack booking platform with room listings and real-time availability checking.
- Implemented end-to-end booking management, from search to confirmation.

LEADERSHIP

IEEE Student Branch, Bannari Amman Institute of Technology, *Vice Secretary*
Coordinating technical events and cross-functional student initiatives alongside outreach efforts for IEEE RAS.

LANGUAGES

English | Tamil